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INFORMATION PROCESSING SYSTEM, INFORMATION PROCESSING METHOD, AND NON-TRANSITORY RECORDING MEDIUM

Abstract

An information processing system includes an information processing apparatus operated by an operator, a function execution apparatus to execute a function, and a management apparatus to communicate with the information processing apparatus and the function execution apparatus. The management apparatus includes first circuitry to generate a code corresponding to a setting to be used in executing the function and transmit the code. The information processing apparatus includes second circuitry to receive the code and display, on a display, the code. The function execution apparatus includes third circuitry to read the code displayed on the display and execute the function based on the setting corresponding to the code read by the third circuitry.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2024-033299, filed on Mar. 5, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to an information processing system, an information processing method, and a non-transitory recording medium.

Related Art

[0003] As known in the art, a cloud print service is used. In the cloud print service, print data is uploaded from, for example, a smartphone to a destination such as a server, and an image forming apparatus such as a multifunction peripheral downloads the print data from the destination and executes printing based on the print data.

[0004] For example, a technique for enabling the cloud print service is disclosed. According to the technique, an operator operates a smartphone to display, on the screen of the smartphone, a QUICK RESPONSE (QR) CODE representing print information including information on a destination and information on print setting. A multifunction peripheral reads and analyzes the QR CODE, decodes the print information encoded in the QR CODE, which includes the information on the destination from which print data is to be acquired and the information on the print setting, and sets the number of sheets to be printed and a sheet size according to the information on the print setting.

SUMMARY

[0005] In one aspect, an information processing system includes an information processing apparatus operated by an operator, a function execution apparatus to execute a function, and a management apparatus to communicate with the information processing apparatus and the function execution apparatus. The management apparatus includes first circuitry to generate a code corresponding to a setting to be used in executing the function and transmit the code.

[0006] The information processing apparatus includes second circuitry to receive the code and display, on a display, the code. The function execution apparatus includes third circuitry to read the code displayed on the display and execute the function based on the setting corresponding to the code read by the third circuitry.

[0007] In another aspect, an information processing system includes an information processing apparatus operated by an operator, a function execution apparatus to execute a function, and a management apparatus to communicate with the information processing apparatus and the function execution apparatus. The management apparatus includes first circuitry to transmit a value corresponding to a setting to be used in executing the function. The information processing apparatus includes second circuitry to receive the value, generate a code corresponding to the setting using the value, and display, on a display, the code. The function execution apparatus includes third circuitry to read the code displayed on the display and execute the function based on the setting corresponding to the code read by the third circuitry.

[0008] In another aspect, an information processing method performed by an information processing system including an information processing apparatus operated by an operator, a function execution apparatus, and a management apparatus, includes, with the management apparatus, generating a code corresponding to a setting to be used in executing a function of the

function execution apparatus, transmitting the code from the management apparatus to the information processing apparatus, displaying the code on a display for the information processing apparatus, reading the displayed code with the function execution apparatus, and executing, with the function execution apparatus, the function based on the setting corresponding to the code that is read.

[0009] In another aspect, a non-transitory recording medium stores a plurality of program codes which, when executed by one or more processors on an information processing system including an information processing apparatus operated by an operator, a function execution apparatus, and a management apparatus, causes the one or more processors to perform an information processing method including, with the management apparatus, generating a code corresponding to a setting to be used in executing a function of the function execution apparatus, transmitting the code from the management apparatus to the information processing apparatus, displaying the code on a display for the information processing apparatus, reading the displayed code with the function execution apparatus, and executing, with the function execution apparatus, the function based on the setting corresponding to the code that is read.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0011] FIG. 1 is a diagram illustrating a configuration of an information processing system according to a first embodiment;

[0012] FIG. 2 is a block diagram illustrating a hardware configuration of a management apparatus;

[0013] FIG. 3 is a block diagram illustrating a hardware configuration of an information processing apparatus;

[0014] FIG. 4 is a block diagram illustrating a hardware configuration of a function execution apparatus;

[0015] FIG. 5 is a block diagram illustrating a functional configuration of each apparatus included in an information processing system according to the first embodiment;

[0016] FIG. 6 is a diagram illustrating a table indicating the correspondence between settings and identifications (IDs);

[0017] FIG. 7 is a flowchart of a process to execute a function according to the first embodiment;

[0018] FIG. 8 is a block diagram illustrating a functional configuration of each apparatus included in an information processing system according to a second embodiment;

[0019] FIG. 9 is a sequence diagram illustrating a process to execute a function according to the second embodiment;

[0020] FIG. 10 is a block diagram illustrating a functional configuration of each apparatus included in an information processing system according to a third embodiment;

[0021] FIG. 11 is a sequence diagram illustrating a process to execute a function according to the third embodiment;

[0022] FIG. 12 is a block diagram illustrating a functional configuration of each apparatus included in an information processing system according to a fourth embodiment;

[0023] FIG. 13 is a diagram illustrating a version list, and sets of information on all settings for each version and IDs corresponding to the respective settings;

[0024] FIG. 14 is a sequence diagram illustrating a process to execute a function according to the fourth embodiment;

[0025] FIG. 15 is a block diagram illustrating a functional configuration of each apparatus included

in an information processing system according to a fifth embodiment; and

[0026] FIG. **16** is a sequence diagram illustrating a process to execute a function according to the fifth embodiment.

[0027] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

DETAILED DESCRIPTION

[0028] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0029] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0030] An information processing system, an information processing method, and a non-transitory recording medium according to embodiments of the present disclosure are described in detail below with reference to the accompanying drawings.

First Embodiment

[0031] FIG. **1** is a diagram illustrating a configuration of an information processing system **1** according to a first embodiment. The information processing system **1** according to the present embodiment includes an information processing apparatus **20**, a function execution apparatus **30**, and a management apparatus **10**, which are connected to one another via a communication network **40** such as the Internet. The information processing apparatus is used by an operator **2**. The function execution apparatus **30** is an example of a multifunction peripheral/printer (MFP) or a facsimile machine. Each apparatus is described in detail below.

[0032] The information processing apparatus **20** is a communication terminal used by the operator **2** to instruct the function execution apparatus **30** to execute a function such as printing or scanning. The information processing apparatus **20** is, for example, a smartphone or a tablet terminal. The function execution apparatus **30** executes a function instructed by the operator **2**. Examples of the function are printing, scanning, copying, and facsimile communication. The management apparatus **10** is a computer connected to a cloud service via the communication network **40**. The management apparatus **10** is an information processing apparatus such as a server, a desktop personal computer (PC), or a laptop personal computer (PC). The management apparatus **10** exchanges information with the information processing apparatus **20** and the function execution apparatus **30**, and provides services for the function execution apparatus **30** to execute the function.

[0033] FIG. **2** is a block diagram illustrating a hardware configuration of the management apparatus **10** implemented by a server, a desktop PC, or a laptop PC.

[0034] As illustrated in FIG. **2**, the management apparatus **10** is implemented by a computer. The computer includes a central processing unit (CPU) **1001**, a read-only memory (ROM) **1002**, a random-access memory (RAM) **1003**, a hard disk (HD) **1004**, a hard disk drive (HDD) controller **1005**, a display **1006**, an external device interface (I/F) **1008**, a network interface (I/F) **1009**, a bus line **1010**, a keyboard **1011**, a pointing device **1012**, a digital versatile disc-rewritable (DVD-RW) drive **1014**, and a medium interface (I/F) **1016**.

[0035] The CPU **1001** controls the entire operation of the management apparatus **10**. The ROM **1002** stores a program such as an initial program loader (IPL) used for booting the CPU **1001**. The RAM **1003** is used as a work area for the CPU **1001**. The HD **1004** is a storage unit that stores various data such as a program. The HDD controller **1005** controls the reading and writing of data from and to the HD **1004** under the control of the CPU **1001**. The display **1006** is a display device

including a liquid crystal panel or an organic electro-luminescence (EL) panel that displays various types of information such as a cursor, a menu, a window, characters, and an image.

[0036] The external device I/F **1008**, which may be implemented by an interface circuit, is an interface for connection with various external devices. The network I/F **1009**, which may be implemented by an interface circuit, is an interface that controls the communication of data through the communication network **40**. The bus line **1010** is, for example, an address bus or a data bus, which electrically connects the components or elements such as the CPU **1001** illustrated in FIG. **2**. [0037] The keyboard **1011** serves as an input device provided with a plurality of keys used for, for example, inputting characters, numerical values, and various instructions. The pointing device **1012** serves as an input device used for, for example, selecting or executing various instructions, selecting an object to be processed, and moving a cursor being displayed. As the pointing device **1012**, a mouse, a touch pad, a trackball, a joystick, or a pen tablet may be used. The display **1006** and the pointing device **1012** may be implemented by a touch panel display having respective functions.

[0038] The DVD-RW drive **1014** controls the reading and writing of various data from and to a DVD-RW **1013**, which serves as a removable recording medium according to the present embodiment. The removable recording medium is not limited to a DVD-RW and may be, for example, a digital versatile disc-recordable (DVD-R). The medium I/F **1016**, which may be implemented by an interface circuit, controls the reading and writing (storing) of data from and to a recording medium **1015** such as a flash memory. The DVD-RW **1013** and the recording medium **1015** are examples of a storage unit for the management apparatus **10**.

[0039] FIG. **3** is a block diagram illustrating a hardware configuration of the information processing apparatus **20** implemented by a smartphone or a tablet terminal. As illustrated in FIG. **3**, the information processing apparatus **20** includes a CPU **2001**, a ROM **2002**, a RAM **2003**, an electrically erasable programmable read-only memory (EEPROM) **2004**, a complementary metal oxide semiconductor (CMOS) sensor **2005**, an imaging element interface (I/F) **2006**, an acceleration and orientation sensor **2007**, a medium I/F **2009**, and a global positioning system (GPS) receiver **2011**.

[0040] The CPU **2001** controls the entire operation of the information processing apparatus **20**. The ROM **2002** is a storage unit that stores, for example, a program used by the CPU **2001** and a program such as an IPL used for booting the CPU **2001**. The RAM **2003** is used as a work area for the CPU **2001**. The EEPROM **2004** is a storage unit that reads and writes various data such as a program for a smartphone under the control of the CPU **2001**. The CMOS sensor **2005** is a kind of built-in imaging device that captures an image of an object (typically, a self-image of the operator **2**) under the control of the CPU **2001** to obtain image data. In alternative to the CMOS sensor, an imaging device such as a charge-coupled device (CCD) sensor may be used. The imaging element I/F **2006** is a circuit that controls the driving of the CMOS sensor **2005**. The acceleration and orientation sensor **2007** serves as various sensors such as an electromagnetic compass or gyrocompass for detecting geomagnetism and an acceleration sensor. The medium I/F **2009**, which may be implemented by an interface circuit, controls the reading and writing (storing) of data from and to a recording medium **2008** such as a flash memory. The recording medium **2008** is an example of a storage unit for the information processing apparatus **20**. The GPS receiver **2011** receives a GPS signal from a GPS satellite.

[0041] The information processing apparatus **20** further includes a long-range communication circuit **2012**, an antenna **2012a** for the long-range communication circuit **2012**, a CMOS sensor **2013**, an imaging element I/F **2014**, a microphone **2015**, a speaker **2016**, an audio input and output interface (I/F) **2017**, a display **2018**, an external device I/F **2019**, a short-range communication circuit **2020**, an antenna **2020a** for the short-range communication circuit **2020**, and a touch panel **2021**.

[0042] The long-range communication circuit **2012** is a circuit for communicating with other

devices via the communication network **40**. The CMOS sensor **2013** is a kind of built-in imaging device that captures an image of an object under the control of the CPU **2001** to obtain image data. The imaging element I/F **2014** is a circuit that controls the driving of the CMOS sensor **2013**.

[0043] The microphone **2015** is a built-in circuit that converts audio into electrical signals. The speaker **2016** is a built-in circuit that converts electrical signals into physical vibrations to produce audio such as music or voice. The audio input and output I/F **2017** is a circuit that processes the input and output of audio signals between the microphone **2015** and the speaker **2016** under the control of the CPU **2001**. The display **2018** serves as a display that displays an image of an object and various icons, such as a liquid crystal display or an organic electro-luminescence (EL) display. The external device I/F **2019**, which may be implemented by an interface circuit, is an interface for connection with various external devices. The short-range communication circuit **2020** is a communication circuit in compliance with, for example, the near field communication (NFC) or BLUETOOTH. The touch panel **2021** is a kind of a pointing device. The touch panel **2021** serves as an input device that allows the operator **2** to operate the information processing apparatus **20** by touching, for example, a button displayed on the display **2018**.

[0044] The information processing apparatus **20** further includes a bus line **2010**. The bus line **2010** is, for example, an address bus or a data bus, which electrically connects the components or elements such as the CPU **2001** illustrated in FIG. **3**.

[0045] FIG. **4** is a block diagram illustrating a hardware configuration of the function execution apparatus **30** implemented by an MFP. As illustrated in FIG. **4**, the function execution apparatus **30** includes a controller **3010**, a short-range communication circuit **3020**, an engine controller **3030**, an operation panel **3040**, a network I/F **3050**, and a code reader **3060**.

[0046] The controller **3010** includes a CPU **3001** as a main processor of the computer, a system memory **3002**, a north bridge (NB) **3003**, a south bridge (SB) **3004**, an application-specific integrated circuit (ASIC) **3006**, a local memory **3007** as a storage unit, an HDD controller **3008**, and an HD **3009** as a storage unit. The NB **3003** and the ASIC **3006** are connected to each other through an accelerated graphics port (AGP) bus **3021**.

[0047] The CPU **3001** controls the entire operation of the function execution apparatus **30**. The NB **3003** is a bridge that connects the CPU **3001** to the system memory **3002**, the SB **3004**, and the AGP bus **3021**. The NB **3003** includes a memory controller to control the reading from and writing to the system memory **3002**, a peripheral component interconnect (PCI) master, and an AGP target.

[0048] The system memory **3002** includes a ROM **3002a** and a RAM **3002b**. The ROM **3002a** stores programs and data for implementing various functions of the controller **3010**. The RAM **3002b** is used as a memory to load the programs and the data. The RAM **3002b** is used as, for example, a drawing memory to store drawing data for printing. For the distribution, the programs stored in the ROM **3002a** may be stored in any computer-readable recording medium, such as a compact disc-read only memory (CD-ROM), a compact disc-recordable (CD-R), or a digital versatile disc (DVD), in a file format installable or executable by a computer.

[0049] The SB **3004** is a bridge for connecting the NB **3003** to PCI devices and peripheral devices. The ASIC **3006** is an integrated circuit (IC) dedicated to image processing and includes hardware elements for image processing. The ASIC **3006** serves as a bridge to connect the AGP bus **3021**, a PCI bus **3022**, the HDD controller **3008**, and the local memory **3007** to one another. The ASIC **3006** includes a PCI target, an AGP master, an arbiter (ARB) as a central processor of the ASIC **3006**, a memory controller to control the local memory **3007**, a plurality of direct memory access controllers (DMACs), and a PCI unit. For example, the DMACs convert coordinates of image data with hardware logic to rotate an image based on the image data. The PCI unit transfers data between a scanner engine **3031** and a printer engine **3032** through the PCI bus **3022**. The ASIC **3006** may be connected to a universal serial bus (USB) interface, or an Institute of Electrical and Electronics Engineers (IEEE) 1394 interface.

[0050] The local memory **3007** is a memory used as a buffer for image data to be copied or a code

buffer. The HD **3009** is a storage device (storage unit) that stores image data, font data for printing, and form data. The HDD controller **3008** controls the reading and writing of data from and to the HD **3009** under the control of the CPU **3001**. The AGP bus **3021** is a bus interface for a graphics accelerator card, which has been proposed to accelerate graphics processing. The AGP bus **3021** directly accesses the system memory **3002** with high throughput to accelerate the graphics accelerator card.

[0051] The short-range communication circuit **3020** includes an antenna **3020a** for the short-range communication circuit **3020**. The short-range communication circuit **3020** is a communication circuit in compliance with, for example, the NFC or BLUETOOTH.

[0052] The engine controller **3030** includes the scanner engine **3031** and the printer engine **3032**. The operation panel **3040** includes a display panel **3040a** and a control panel **3040b**. The display panel **3040a** is, for example, a touch panel that displays current settings or a selection screen and receives input from the operator **2**. The control panel **3040b** includes, for example, a numeric keypad and a start key. The numeric keypad receives assigned values of image reading or forming parameters such as an image density parameter. The start key receives an instruction to start copying, for example. The controller **3010** controls the entire operation of the function execution apparatus **30**. For example, the controller **3010** controls drawing, communication, and input through the operation panel **3040**. The scanner engine **3031** or the printer engine **3032** includes an image processor to perform, for example, error diffusion and gamma conversion.

[0053] In response to an instruction to select a specific application through the operation panel **3040** by use of, for example, an application (or mode) switch key, the function execution apparatus **30** selectively performs a document server function, a copying function, a printing function, and a facsimile communication function. When the document server function is selected, an operation mode switches to a document box mode. With selection of the copying function, the operation mode switches to a copy mode. With selection of the printing function, the operation mode switches to a printer mode. With selection of the facsimile communication function, the operation mode switches to a facsimile mode.

[0054] The network I/F **3050**, which may be implemented by an interface circuit, is an interface that controls the communication of data through the communication network **40**. The short-range communication circuit **3020** and the network I/F **3050** are electrically connected to the ASIC **3006** via the PCI bus **3022**.

[0055] The code reader **3060** is a code reading device that reads a code displayed on the display of the information processing apparatus **20** as described below. In the case where the code is an image representing characters such as numerals, the code reader includes a camera and recognizes the numerals captured by the camera using a technique of image recognition.

[0056] In the case where the code is a bar code or a QUICK RESPONSE (QR) CODE, the code reader serves as a bar code reader or a QR code reader, and outputs information embedded in the code read by the code reader. For example, the code reader may be connected to the main body of the function execution apparatus **30** through a cable, or may be provided for the operation panel **3040**.

[0057] FIG. **5** is a block diagram illustrating a functional configuration of each apparatus included in the information processing system **1** according to the first embodiment.

[0058] The management apparatus **10** includes a transmission unit **111** and a code generation unit **112**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the management apparatus **10** as illustrated in FIG. **2**. Alternatively, at least one of the functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the management apparatus **10** are not limited thereto.

[0059] The information processing apparatus **20** includes a reception unit **211** and a code display unit **212**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the information processing apparatus **20** as illustrated in FIG. **3**.

Alternatively, at least one of the functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the information processing apparatus **20** are not limited thereto.

[0060] The function execution apparatus **30** includes a code reading unit **311** and a function execution unit **312**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the function execution apparatus **30** as illustrated in FIG. **4**. Alternatively, at least one of the functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the function execution apparatus **30** are not limited thereto.

[0061] The transmission unit **111** transmits a code corresponding to a setting registered by the operator **2** to the information processing apparatus **20**. In the present embodiment, the setting is used for executing a function such as printing or scanning. Examples of the setting include a setting of color printing or monochrome printing, a setting of single-sided printing or double-sided printing, a setting of a sheet size, and a setting of resolution in scanning. It is assumed that the operator **2** registers a setting of a function in the information processing apparatus **20** using an input device such as the touch panel **2021** and the information processing apparatus **20** transmits the information on the setting to the management apparatus **10**.

[0062] The code generation unit **112** generates a code corresponding to the setting registered by the operator **2**. In the present embodiment, the code is information that allows all settings (all combinations of setting values) that can be set for a function to be executed to be identified. The code may be an identification (ID) (e.g., a value) such as a numerical value corresponding to each setting, or may be a one-dimensional code such as a bar code or a two-dimensional code such as a QR CODE. In the case where the code is a one-dimensional code or a two-dimensional code, a code may be generated in which an ID corresponding to the setting is embedded.

[0063] FIG. **6** is a diagram illustrating a table indicating the correspondence between settings and IDs. In the table, the IDs are indicated in the leftmost column, and the settings (a specific setting value of each setting item) corresponding to the respective IDs are indicated in the columns to the right of the IDs. The management apparatus **10** stores information on all settings (setting information) and IDs corresponding to the respective settings as sets in a storage unit such as the HD **1004**.

[0064] The reception unit **211** receives the code transmitted from the transmission unit **111**. The code display unit **212** displays the code received by the reception unit **211** on a display such as the display **2018**. The code display unit **212** displays a numerical value when the code is a numerical value such as an ID, and displays an image of a two-dimensional code when the code is a QR CODE.

[0065] The code reading unit **311** reads the code displayed by the code display unit **212**. The code reading unit **311** reads the numerical value in the code using a technique of image recognition or character recognition when the code is a numerical value, and reads the code using a bar code reader or a QR code reader when the code is a bar code or a QR CODE.

[0066] The function execution unit **312** executes a function such as printing or scanning based on a setting corresponding to the code read by the code reading unit **311**. The function execution apparatus **30** stores the information on all settings (setting information) and the IDs corresponding to the respective settings as sets in a storage unit such as the HD **3009**. In the case where the code is a numerical value such as an ID, the function execution unit **312** searches the table illustrated in FIG. **6** to acquire a setting corresponding to the ID, and executes a function based on the setting. In the case where the code is a QR CODE, the code reading unit **311** acquires an ID embedded in the QR CODE, and the function execution unit **312** acquires a setting corresponding to the ID using the ID received from the code reading unit **311** and the table.

[0067] FIG. **7** is a flowchart of a process to execute a function according to the first embodiment. First, in step **S10**, the code generation unit **112** generates a code corresponding to a setting to be registered by the operator **2**. The operator **2** operates an input device such as the touch panel **2021** of the information processing apparatus **20** to select a setting to be registered, and presses a

registration button displayed on the display **2018** to register the setting. The information on the registered setting is transmitted from the information processing apparatus **20** to the management apparatus **10** via the communication network **40**.

[0068] In step **S11**, the transmission unit **111** transmits a code corresponding to the setting registered by the operator **2** to the information processing apparatus **20**. In step **S12**, the reception unit **211** receives the code transmitted from the transmission unit **111**. In step **S13**, the code display unit **212** displays the code received by the reception unit **211** on a display such as the display **2018**.

[0069] In step **S14**, the operator **2** holds the code displayed on the code display unit **212** over the code reading unit **311**, and the code reading unit **311** reads the displayed code. In step **S15**, the function execution unit **312** executes a function based on the setting corresponding to the read code.

[0070] According to the technique in the art, since the amount of data of the information on the print setting is large, there are some issues that it takes time to download the information on the print setting in a network of slow speed, and that the storage capacity of the server is significantly affected when the information on the print setting is stored for operators individually. In addition, since the target of the cloud printing service is limited to the printing function, there is another issue that the cloud printing service cannot be used for services to executes a variety of functions including functions other than the printing function.

[0071] As described above, according to the present embodiment, the setting information is transmitted to the function execution apparatus **30** by holding the code displayed by the information processing apparatus **20** over the function execution apparatus **30** that can execute various functions. Accordingly, even when the amount of data in the setting information is large, the setting information registered by the operator **2** is smoothly transmitted to the function execution apparatus **30**. In addition, the management apparatus **10** generates a code corresponding to the setting. Thus, the information processing apparatus **20** does not need to generate the code, and the amount of processing executed by the information processing apparatus **20** is reduced. Furthermore, the management apparatus **10** and the function execution apparatus **30** store, in the respective storage units, the information on the settings having a finite number of combinations of setting values and the IDs corresponding to the respective settings. Thus, the amount of data to be stored is reduced, compared to, for example, a case where the information on the settings is stored for an unspecified number of operators individually.

[0072] In the present embodiment, the management apparatus **10** includes the code generation unit. Alternatively, the information processing apparatus **20** may include the code generation unit, and the information processing apparatus **20** executes the processing to generate a code based on the setting registered by the operator **2**. In this case, the ID (value) corresponding to the setting registered by the operator **2** may be transmitted from the transmission unit **111** to the reception unit **211**, and the information processing apparatus **20** may generate a code (a code in which the ID is embedded) corresponding to the received ID. Thus, the data transmitted from the management apparatus **10** to the information processing apparatus **20** is data such as an ID whose amount of information is small, and the amount of data related to the setting information transmitted from the management apparatus **10** to the information processing apparatus **20** is reduced.

Second Embodiment

[0073] A second embodiment is described below. In the present embodiment, how data such as setting information and an ID is exchanged between the apparatuses to execute a function is described in more detail.

[0074] FIG. **8** is a block diagram illustrating a functional configuration of each apparatus included in the information processing system **1** according to the second embodiment. The management apparatus **10** includes a first setting control unit **121** and a first setting storage unit **122**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the management apparatus **10** as illustrated in FIG. **2**. Alternatively, at least one of the

functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the management apparatus **10** are not limited thereto.

[0075] The information processing apparatus **20** includes a second setting control unit **221**, a second setting storage unit **222**, and a second code generation unit **223**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the information processing apparatus **20** as illustrated in FIG. **3**. Alternatively, at least one of the functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the information processing apparatus **20** are not limited thereto.

[0076] The function execution apparatus **30** includes a third setting control unit **321**, a third setting storage unit **322**, a code reading unit **323**, and a function execution unit **324**. These functional units are implemented by cooperation of the hardware and software (e.g., a program) included in the function execution apparatus **30** as illustrated in FIG. **4**. Alternatively, at least one of the functional units may be implemented by dedicated hardware (e.g., a circuit). The functional units of the function execution apparatus **30** are not limited thereto.

[0077] As described above, the management apparatus **10** stores the information on all settings and the IDs corresponding to the respective settings as sets in a storage unit such as the HD **1004**.

[0078] The first setting control unit **121** reads the information on all settings and the IDs from the first setting storage unit **122**, and transmits the read data to the function execution apparatus **30**. In addition, the first setting control unit **121** inquires of the first setting storage unit **122** for the ID (value) corresponding to a specific setting information (information on the setting registered by the operator **2**) received from the information processing apparatus **20**, and transmits the ID (value) returned from the first setting storage unit **122** to the information processing apparatus **20**. As described above, the first setting control unit **121** has a transmission function that is a function to transmit a value corresponding to the setting.

[0079] The first setting storage unit **122** reads the information on all settings and the IDs from the storage unit and transmits the read data to the first setting control unit **121**. The first setting storage unit **122** returns an ID corresponding to the specific setting transmitted from the first setting control unit **121** to the first setting control unit **121**. In the present embodiment, the first setting storage unit **122** is an example of a storage unit that reads or writes setting information and an ID.

[0080] The information on the setting registered by the operator **2** and the code corresponding to the setting are stored in the storage unit such as the EEPROM **2004** of the information processing apparatus **20**. Multiple sets of setting information and codes may be stored in the storage unit.

[0081] The second setting control unit **221** transmits the information on the setting registered by the operator **2** to the management apparatus **10**, and receives an ID (a value) corresponding to the setting from the first setting control unit **121**. As described above, the second setting control unit **221** has a reception function that is a function to receive a value corresponding to the setting.

[0082] The second setting control unit **221** requests the second code generation unit **223** to generate a code corresponding to the received ID, instructs the second setting storage unit **222** to store the information on the setting registered by the operator **2** and the code as a set, and displays the setting registered by the operator **2** and the code on a display such as the display **2018**. As described above, the second setting control unit **221** has a function similar to the function of the code display unit **212** described in the first embodiment.

[0083] The second setting storage unit **222** stores the information on the setting registered by the operator **2** and the code as a set in the storage unit. The second code generation unit **223** generates a code corresponding to the ID. In the present embodiment, the second setting storage unit **222** is an example of a storage unit that reads or writes setting information and an ID.

[0084] The information on all settings and the IDs corresponding to the respective settings transmitted from the management apparatus **10** are stored as sets in a storage unit such as the HD **3009** of the function execution apparatus **30**. For this purpose, the third setting control unit **321** requests the first setting control unit **121** to transmit the information on all settings and the IDs. The

third setting control unit **321** requests the third setting storage unit **322** to transmit the setting information corresponding to the ID transmitted from the code reading unit **323**, and returns the setting information corresponding to the ID received from the third setting storage unit **322** to the function execution unit **324** in response to the request for the setting information from the function execution unit **324**.

[0085] The third setting storage unit **322** reads the setting information corresponding to the ID received from the third setting control unit **321** from the storage unit and returns the setting information to the third setting control unit **321**. The code reading unit **323** transmits an ID corresponding to the read code to the third setting control unit **321**. In the present embodiment, the third setting storage unit **322** is an example of a storage unit that reads or writes setting information and an ID.

[0086] FIG. **9** is a sequence diagram illustrating a process to execute a function according to the second embodiment. In the present embodiment, a solid arrow indicates, for example, a request for processing or operation, an instruction, or data registration. In the case where a solid arrow indicates a request or instruction, a dotted arrow indicates a response corresponding to the request or instruction.

[0087] First, in step **S201**, the third setting control unit **321** requests the first setting control unit **121** to transmit the information on all settings and the IDs. In step **S202**, the first setting control unit **121** requests the first setting storage unit **122** to read out the information on all settings and the IDs. In step **S203**, the first setting storage unit **122** reads the information on all settings and the IDs from the storage unit. In steps **S204** and **S205**, the first setting control unit **121** transmits the information on all settings and the IDs received from the first setting storage unit **122** to the third setting control unit **321**. The operations described above are executed in advance before the operator **2** uses the function execution apparatus **30**. For example, at a timing when the operator **2** is not using the function execution apparatus **30**, the function execution apparatus **30** requests the management apparatus **10** for the setting information.

[0088] In step **S206**, the operator **2** registers a favorite setting or a frequently used setting using the information processing apparatus **20**. For example, the operator **2** operates an input device such as the touch panel **2021** of the information processing apparatus **20** to select a setting to be registered, and presses a registration button displayed on the display **2018** to register the setting. In step **S207**, the second setting control unit **221** transmits the information on the setting registered in the information processing apparatus **20** to the first setting control unit **121**.

[0089] In step **S208**, the first setting control unit **121** requests the first setting storage unit **122** to transmit an ID corresponding to the received setting information. In step **S209**, the first setting storage unit **122** searches the storage unit for the requested ID. In step **S210**, the first setting storage unit **122** transmits the searched ID to the first setting control unit **121**. In step **S211**, the first setting control unit **121** transmits the received ID to the second setting control unit **221**.

[0090] In step **S212**, the second setting control unit **221** requests the second code generation unit **223** to generate a code corresponding to the received ID. In step **S213**, the second code generation unit **223** generates a code corresponding to the ID transmitted from the second setting control unit **221**. In step **S214**, the second code generation unit **223** returns the generated code to the second setting control unit **221**. In step **S215**, the second setting control unit **221** requests the second setting storage unit **222** to store the setting information and the code corresponding to the setting information. In step **S216**, the second setting storage unit **222** stores the setting information and the code corresponding to the setting information. In step **S217**, the second setting storage unit **222** notifies the second setting control unit **221** of the completion of the storage. In steps **S218** and **S219**, the second setting control unit **221** displays the setting information and the code corresponding to the setting information on the display.

[0091] In step **S220**, the operator **2** holds the code displayed on the display of the information processing apparatus **20** over the code reading unit **323** to cause the code reading unit **323** to read

the code.

[0092] In step S221, the code reading unit **323** transmits an ID corresponding to the read code to the third setting control unit **321**. In step S222, the third setting control unit **321** causes the code reading unit **323** to end the reading operation. In step S223, the third setting control unit **321** requests the third setting storage unit **322** to transmit setting information corresponding to the received ID. In step S224, the third setting storage unit **322** searches the storage unit for the setting information corresponding to the ID. In step S225, the third setting storage unit **322** transmits the searched setting information to the third setting control unit **321**.

[0093] In step S226, the operator **2** presses an execution button displayed on, for example, the display **2018** of the function execution apparatus **30**. In step S227, the function execution unit **324** inquires of the third setting control unit **321** for the setting information. In step S228, the third setting control unit **321** transmits the setting information received from the third setting storage unit **322** in step S225 to the function execution unit **324**. In steps S229 and S230, the function execution unit **324** executes a function using a setting corresponding to the received setting information. The sequence when the operator **2** registers one setting is described above. By repeating the operations in steps S206 to S219, the operator **2** is allowed to register multiple settings.

[0094] As described above, according to the present embodiment, by using the ID corresponding to the information on the setting registered by the operator **2**, the amount of data of the setting information transmitted from the management apparatus **10** to the information processing apparatus **20** is reduced. In addition, the setting information is transmitted to the function execution apparatus **30** by holding the code generated by the information processing apparatus **20** over the function execution apparatus **30** that can execute various functions. Accordingly, even when the amount of data in the setting information is large, the setting information registered by the operator **2** is smoothly transmitted to the function execution apparatus **30**. Furthermore, the management apparatus **10** and the function execution apparatus **30** store, in the respective storage units, the information on the settings having a finite number of combinations of setting values and the IDs corresponding to the respective settings. Thus, the amount of data to be stored is reduced, compared to, for example, a case where the information on the settings is stored for an unspecified number of operators individually.

Third Embodiment

[0095] A third embodiment is described below. In the present embodiment, instead of the information processing apparatus **20**, the management apparatus **10** has the function to generate a code.

[0096] FIG. **10** is a block diagram illustrating a functional configuration of each apparatus included in the information processing system **1** according to the third embodiment. The difference from the second embodiment is that the management apparatus **10** includes a first code generation unit **133** instead of the information processing apparatus **20** including the second code generation unit **223**. Accordingly, a code corresponding to the setting information can be generated by the management apparatus **10** and transmitted to the information processing apparatus **20**, as described later. In the following description of the third embodiment, descriptions of elements identical or similar to those in the second embodiment are omitted, and differences from the second embodiment are described.

[0097] The first setting control unit **131** requests the first code generation unit **133** to generate a code corresponding to the ID corresponding to the setting registered by the operator **2**. The first code generation unit **133** generates a code corresponding to the ID. The first setting control unit **131** transmits the generated code to the second setting control unit **231**. As described above, the first setting control unit **131** has a function similar to the function of the transmission unit **111** described in the first embodiment.

[0098] The second setting control unit **231** receives the code transmitted from the first setting control unit **131**. As described above, the second setting control unit **231** has a function similar to

the function of the reception unit **211** described in the first embodiment.

[0099] The storage unit of the information processing apparatus **20** stores the information on the setting registered by the operator **2** and the code corresponding to the setting. Multiple sets of setting information and codes may be stored in the storage unit.

[0100] FIG. **11** is a sequence diagram illustrating a process to execute a function according to the third embodiment. The features different from the second embodiment are described below.

[0101] In step **S311**, the first setting control unit **131** requests the first code generation unit **133** to generate a code corresponding to the ID received in step **S310**. In step **S312**, the first code generation unit **133** generates a code corresponding to the received ID. In step **S313**, the first code generation unit **133** returns the generated code to the first setting control unit **131**. In step **S314**, the first setting control unit **131** transmits the received code to the second setting control unit **231**. In steps **S315** and **S316**, the second setting control unit **231** requests the second setting storage unit **232** to store the setting information and the ID and code corresponding to the setting information, and the second setting storage unit **232** stores the setting information, the ID, and the code in the storage unit.

[0102] As described above, according to the present embodiment, since the management apparatus **10** executes the function to generate a code corresponding to the ID, the amount of processing executed by the information processing apparatus **20** is reduced. In addition, the setting information is transmitted to the function execution apparatus **30** by holding the code displayed by the information processing apparatus **20** over the function execution apparatus **30** that can execute various functions. Accordingly, even when the amount of data in the setting information is large, the setting information registered by the operator **2** is smoothly transmitted to the function execution apparatus **30**. Furthermore, the management apparatus **10** and the function execution apparatus **30** store, in the respective storage units, the information on the settings having a finite number of combinations of setting values and the IDs corresponding to the respective settings. Thus, the amount of data to be stored is reduced, compared to, for example, a case where the information on the settings is stored for an unspecified number of operators individually.

Fourth Embodiment

[0103] When a function executed by the function execution apparatus **30** includes multiple versions, a setting item for the function of a certain version may not be supported by the function of another version. For example, the function of version A does not include a setting item for double-sided printing, whereas the function of updated version B includes a setting item newly added for double-sided printing. In a fourth embodiment, even when the version corresponding to the setting registered by the information processing apparatus **20** is different from the version that the function execution apparatus **30** supports, the function can be executed.

[0104] FIG. **12** is a block diagram illustrating a functional configuration of each apparatus included in the information processing system **1** according to the fourth embodiment. The difference from the second embodiment is that the first setting storage unit **142** includes a first version storage unit **1421**, and the third setting storage unit **342** include a third version storage unit **3421**. In the following description of the fourth embodiment, descriptions of elements identical or similar to those in the second embodiment are omitted, and differences from the second embodiment are described.

[0105] In the storage unit of the management apparatus **10**, a version list for a function, and the information on all settings for each version and the IDs (values) corresponding to the respective settings as sets are stored. The first setting control unit **141** reads the version list, the information on all settings for each version, and the IDs from the first setting storage unit **142**, and transmits the read data to the function execution apparatus **30**. The first setting control unit **141** inquires of the first setting storage unit **142** for a value of the latest version corresponding to a specific setting information transmitted from the information processing apparatus **20** and an ID corresponding to the specific setting information, and transmits the value and the ID returned from the first setting

storage unit **142** to the information processing apparatus **20**.

[0106] FIG. **13** is a diagram illustrating the version list, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings. On the left section of FIG. **13**, a table of the version list is illustrated. The version list contains the values (one, two, three, and so on) of the respective versions. On the right section of FIG. **13**, tables of the sets of the information on all settings corresponding to each version and the IDs corresponding to the respective setting are illustrated. In the present embodiment, sets of the information on all settings and the IDs corresponding to the respective settings when the value of the version is one, and sets of the information on all settings and the IDs corresponding to the respective settings when the value of the version is two are presented.

[0107] The first setting storage unit **142** reads the version list and the sets of the information on all settings for each version and the IDs corresponding to the respective settings from the storage unit, and returns the read data to the first setting control unit **141**. The version list is read out by the first version storage unit **1421** included in the first setting storage unit **142**. The first setting storage unit **142** returns the value of the latest version and the ID corresponding to a specific setting information inquired by the first setting control unit **141** to the first setting control unit **141**. In the present embodiment, the first setting storage unit **142** is an example of a storage unit that reads or writes setting information and an ID for each version.

[0108] The second setting control unit **241** transmits the setting information registered by the operator **2** to the first setting control unit **141**, receives the value of the version and the ID corresponding to the setting information from the first setting control unit **141**, and requests the second code generation unit **243** to generate a code corresponding to the received value of the version and ID. The second setting control unit **241** instructs the second setting storage unit **242** to store the setting information registered by the operator **2** and the code generated by the second code generation unit **243**, and displays the setting information and the code on the display. As described above, the second setting control unit **241** has a function similar to the function of the code display unit **212** described in the first embodiment. The second setting storage unit **242** stores the setting information registered by the operator **2** and the code generated by the second code generation unit **243** in the storage unit.

[0109] In the storage unit of the function execution apparatus **30**, the version list transmitted from the management apparatus **10**, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings are stored. For this purpose, the third setting control unit **341** requests the first setting control unit **141** to transmit the version list, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings, and instructs the third setting storage unit **342** to store the received data. The version list is stored in the storage unit by the third version storage unit **3421** included in the third setting storage unit **342**. In the present embodiment, the third setting storage unit **342** is an example of a storage unit that reads or writes setting information and an ID for each version.

[0110] The third setting control unit **341** requests the third setting storage unit **342** to transmit setting information corresponding to the value of the version and the ID transmitted from the code reading unit **343**, and checks the setting items that the function execution apparatus **30** supports among the setting items included in the received setting information. When the received setting information includes a setting item that the function execution apparatus **30** does not support, the third setting control unit **341** displays the setting item on the display to present the setting item to the operator **2**. When the function cannot be executed only with the setting items included in the received setting information, the third setting control unit **341** displays the missing setting item on the display to present the missing setting item to the operator **2**. When the operator **2** inputs setting information for an additional setting item through the operation panel **3040**, the third setting control unit **341** merges the setting information input by the operator **2** with the setting information received from the third setting storage unit **342**. By excluding a setting item that the function

execution apparatus **30** does not support and adding a setting item required for the function execution apparatus **30** to execute the function, the third setting control unit **341** merges pieces of the setting information so that there is neither excess nor deficiency in the setting information. [0111] When requested by the function execution unit **344** to transmit the setting information, the third setting control unit **341** returns the setting information corresponding to the value of the version and the ID received from the third setting storage unit **342**. The third setting storage unit **342** reads the setting information corresponding to the value of the version and the ID received from the third setting control unit **341** from the storage unit, and returns the setting information to the third setting control unit **341**.

[0112] The code reading unit **343** transmits the value of the version and the ID corresponding to the read code to the third setting control unit **341**.

[0113] FIG. **14** is a sequence diagram illustrating a process to execute a function according to the fourth embodiment. The features different from the second embodiment are described below.

[0114] First, in step **S401**, the third setting control unit **341** requests the first setting control unit **141** to download a version list, and sets of the information on all settings for each version and the IDs corresponding to the respective settings. In step **S402**, the first setting control unit **141** requests the first setting storage unit **142** to transmit the version list, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings. In step **S403**, the first setting storage unit **142** reads the version list, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings from the storage unit. In steps **S404** and **S405**, the first setting control unit **141** transmits the version list, and the sets of the information on all settings for each version and the IDs corresponding to the respective settings received from the first setting storage unit **142** to the third setting control unit **341**.

[0115] In step **S409**, the first setting storage unit **142** (and the first version storage unit **1421** included in the first setting storage unit **142**) searches for an ID corresponding to the setting information registered by the operator **2** from the setting information of the latest version stored in the storage unit. In step **S410**, the first setting storage unit **142** transmits the ID searched for in step **S409** and the value of the version to the first setting control unit **141**. In step **S411**, the first setting control unit **141** transmits the received ID and value of the version to the second setting control unit **241**.

[0116] In step **S420**, the operator **2** holds the code displayed on the display of the information processing apparatus **20** over the code reading unit **343**. In steps **S421** and **S422**, the code reading unit **343** transmits the value of the version and the ID corresponding to the read code to the third setting control unit **341**.

[0117] In step **S423**, the third setting control unit **341** requests the third setting storage unit **342** (and the third version storage unit **3421** included in the third setting storage unit **342**) to transmit the setting information corresponding to the value of the version and the ID. In step **S424**, the third setting storage unit **342** reads the setting information corresponding to the value of the version and the ID from the data stored in the storage unit. In step **S425**, the third setting storage unit **342** transmits the read setting information to the third setting control unit **341**. In step **S426**, the third setting control unit **341** checks the difference between the setting information received from the third setting storage unit **342** and the setting information that the function execution apparatus **30** supports.

[0118] When the setting information received from the third setting storage unit **342** includes a setting item that the function execution apparatus **30** does not support, in step **S427**, the third setting control unit **341** displays the setting item that cannot be reflected in the execution of the function on the display to presents the setting item to the operator **2**. In step **S428**, the operator **2** checks the setting item that cannot be reflected, and additionally inputs a setting as needed using the operation panel **3040**. In step **S429**, the third setting control unit **341** merges the setting information received from the third setting storage unit **342** and the setting information additionally

input by the operator **2** through the operation panel **3040**.

[0119] In the case where the setting items received from the third setting storage unit **342** are insufficient for the setting and the function cannot be executed, in step **S430**, the third setting control unit **341** displays a message indicating that an additional setting is required on the display to present the message to the operator **2**. In step **S431**, the operator **2** checks the message and additionally input a setting using the operation panel **3040** as needed. In step **S432**, the third setting control unit **341** merges the setting received from the setting storage unit and the setting additionally input by the operator **2**.

[0120] The sequence when the operator **2** registers one setting is described above. By repeating the operations in steps **S406** to **S419**, the operator **2** is allowed to register multiple settings.

[0121] As described above, according to the present embodiment, even when the version corresponding to the setting registered by the information processing apparatus **20** is different from the version supported by the function execution apparatus **30** that can execute various functions, the function can be executed. In other words, even when the setting information registered by the operator **2** is not supported by the function execution apparatus **30** or even when the function cannot be executed due to a lack of a setting item, the function can be executed by presenting the required setting information to the operator **2** and additionally receiving the setting information.

Fifth Embodiment

[0122] A fifth embodiment is described below. In the present embodiment, the function to generate a code described in the fourth embodiment is provided not for the information processing apparatus **20** but for the management apparatus **10**.

[0123] FIG. **15** is a block diagram illustrating a functional configuration of each apparatus included in the information processing system **1** according to the fifth embodiment. The difference from the fourth embodiment is that the management apparatus **10** includes a first code generation unit **153** instead of the information processing apparatus **20** including the second code generation unit **243**. Accordingly, as in the third embodiment, a code corresponding to the setting information is generated by the management apparatus **10** and transmitted to the information processing apparatus **20**. The code in the present embodiment corresponds to the value of a specific version and the setting information corresponding to the specific version, as in the fourth embodiment. In the following description of the fifth embodiment, descriptions of elements identical or similar to those in the fourth embodiment are omitted, and differences from the fourth embodiment are described.

[0124] The first setting control unit **151** requests the first code generation unit **153** to generate a code corresponding to the ID and version of the setting registered by the operator **2**, and transmits the generated code to the second setting control unit **251**. The first code generation unit **153** generates a code corresponding to the ID and the version.

[0125] FIG. **16** is a sequence diagram illustrating a process to execute a function according to the fifth embodiment. The features different from the fourth embodiment are described below.

[0126] In step **S511**, the first setting control unit **151** requests the first code generation unit **153** to generate a code corresponding to the ID and version received in step **S510**. In step **S512**, the first code generating unit **153** generates a code corresponding to the received ID and version. In step **S513**, the first code generation unit **153** returns the generated code to the first setting control unit **151**. In step **S514**, the first setting control unit **151** transmits the received code to the second setting control unit **251**. In steps **S515** and **S516**, the second setting control unit **251** requests the second setting storage unit **252** to store the setting information and the ID and code corresponding to the setting information, and the second setting storage unit **252** stores the setting information, the ID, and the code in the storage unit.

[0127] As described above, according to the present embodiment, since the management apparatus **10** executes the function to generate a code corresponding to the value of the version and the ID in the version, the amount of processing executed by the information processing apparatus **20** is reduced. In addition, the setting information is transmitted to the function execution apparatus **30**

by holding the code displayed by the information processing apparatus **20** over the function execution apparatus **30** that can execute various functions. Accordingly, even when the amount of data in the setting information is large, the setting information registered by the operator **2** is smoothly transmitted to the function execution apparatus **30**. Furthermore, the management apparatus **10** and the function execution apparatus **30** store, in the respective storage units, the information on the settings having a finite number of combinations of setting values and the IDs corresponding to the respective settings. Thus, the amount of data to be stored is reduced, compared to, for example, a case where the information on the settings is stored for an unspecified number of operators individually.

[0128] Although the embodiments of the present disclosure have been described above, the above-described embodiments are presented as examples and are not intended to limit the scope of the present disclosure. Such novel embodiments may be carried out in various other modified forms. Various omissions, substitutions, and changes may be made without departing from the gist of the present disclosure. Such novel embodiments and modifications thereof are within the scope and gist of the present disclosure and are also within the scope of the claims and the equivalent thereof. The elements of different embodiments or modifications may be combined with each other as appropriate.

[0129] For example, in the first to fifth embodiments, the function execution apparatus **30** may include a code generation unit and a code display unit, generates a code corresponding to the setting used for the function to be executed, and displays the code on the display. The information processing apparatus **20** may include a code reading unit and reads the code displayed on the display of the function execution apparatus **30**. In this case, the information processing apparatus **20** includes a transmission unit that transmits the ID corresponding to the read code to the management apparatus **10**, and the management apparatus **10** transmits the setting information corresponding to the received ID to the information processing apparatus **20**. By storing the setting information received by the information processing apparatus **20** and the read code in the storage unit of the information processing apparatus **20**, the time and effort required for the operator in manually registering the setting information is eliminated. Thus, the convenience is increased for the operator.

[0130] Each of the functions of the embodiments described above may be implemented by one or more processing circuits or circuitry. The “processing circuit or circuitry” herein includes a programmed processor to execute each function by software, such as a processor implemented by an electronic circuit, and devices, such as an application-specific integrated circuit (ASIC), a digital signal processor (DSP), a field-programmable gate array (FPGA), and circuit modules known in the art arranged to perform the recited functions.

[0131] The group of apparatuses or devices described in the above-described embodiments of the present disclosure are merely one example of a plurality of computing environments that implement embodiments of the present disclosure. In some embodiments, each apparatus or device may include multiple computing devices such as a server cluster. The multiple computing devices communicate with one another through any type of communication link including, for example, a network or a shared memory, and perform the operations disclosed herein.

[0132] Aspects of the present disclosure are, for example, as follows.

[0133] According to Aspect 1, an information processing system includes an information processing apparatus operated by an operator, a function execution apparatus that executes a function in a setting, and a management apparatus that communicates with the information processing apparatus and the function execution apparatus. The management apparatus includes a code generation unit that generates a code corresponding to the setting, and a transmission unit that transmits the code. The information processing apparatus includes a reception unit that receives the code, and a code display unit that displays the code. The function execution apparatus includes a code reading unit that reads the code displayed by the code display unit, and a function execution

unit that executes the function. The function execution unit executes the function based on the setting corresponding to the code read by the code reading unit.

[0134] According to Aspect 2, an information processing system includes an information processing apparatus operated by an operator, a function execution apparatus that executes a function in a setting, and a management apparatus that communicates with the information processing apparatus and the function execution apparatus. The management apparatus includes a transmission unit that transmits a value corresponding to the setting. The information processing apparatus includes a reception unit that receives the value corresponding to the setting, a code generation unit that generates a code corresponding to the setting, and a code display unit that displays the code. The function execution apparatus includes a code reading unit that reads the code displayed by the code display unit and a function execution unit that executes the function. The function execution unit executes the function based on the setting corresponding to the code read by the code reading unit.

[0135] According to Aspect 3, in the information processing system of Aspect 2, the management apparatus further includes a storage unit that stores information on all settings used for executing the function and codes corresponding to the respective settings, the information processing apparatus further includes a storage unit that stores the value corresponding to the setting received by the reception unit and a setting corresponding to the code, and the function execution apparatus further includes a storage unit that stores the information on all settings and the codes corresponding to the respective settings.

[0136] According to Aspect 4, in the information processing system of Aspect 1, the management apparatus further includes a storage unit that stores information on all settings used for executing the function and codes corresponding to the respective settings, the information processing apparatus further includes a storage unit that stores the code received by the reception unit and the setting corresponding to the code, and the function execution apparatus further includes a storage unit that stores the information on all settings and the codes corresponding to the respective settings.

[0137] According to Aspect 5, in the information processing system of any one of Aspects 1 to 4, the function includes multiple versions, the management apparatus further includes a storage unit that stores all settings used for each of the multiple versions and codes corresponding to the respective settings for each version, the code generation unit generates a code corresponding to one of the versions and one of the settings, the function execution apparatus further includes a storage unit that stores all the settings used for each of the multiple versions and the codes corresponding to the respective settings for each version, and the function execution apparatus executes the function based on the one of the versions and the one of the settings corresponding to the code.

[0138] According to Aspect 6, in the information processing system of any one of Aspects 1 to 5, the function execution apparatus further includes a code generation unit that generates a code corresponding to the setting, and a code display unit that display the code, the information processing apparatus further includes a code reading unit that reads the code displayed by the code display unit of the function execution apparatus, a transmission unit that transmits a value corresponding to the code read by the code reading unit, and a storage unit that stores the setting and the code corresponding to the setting, and the management apparatus further includes a reception unit that receives the value transmitted by the transmitting unit of the information processing apparatus. The management apparatus transmits information on a setting corresponding to the value. The information processing apparatus stores the setting transmitted by the management apparatus and the code read by the code reading unit of the information processing apparatus.

[0139] According to Aspect 7, in the information processing system of any one of Aspects 1 to 6, a value different from each other is assigned to the code for each of all the settings used for executing the function.

[0140] According to Aspect 8, in the information processing system of any one of Aspects 1 to 7, the code includes a two-dimensional code.

[0141] According to Aspect 9, in the information processing system of any one of Aspects 1 to 8, the function execution apparatus is an image forming apparatus.

[0142] According to Aspect 10, an information processing method is used by an information processing apparatus operated by an operator, a function execution apparatus that executes a function in a setting, and a management apparatus that communicates with the information processing apparatus and the function execution apparatus. The method includes generating a code corresponding to the setting with the management apparatus, transmitting the code with the management apparatus, receiving the code with the information processing apparatus, displaying the code with the information processing apparatus, reading the displayed code with the function execution apparatus, and executing the function with the function execution apparatus. The method further includes executing, with the function execution apparatus, the function based on a setting corresponding to the code read with the function execution apparatus.

[0143] According to Aspect 11, a non-transitory recording medium stores a plurality of program codes which, when executed by one or more processors, causes the one or more processors to perform a method used by an information processing apparatus operated by an operator, a function execution apparatus that executes a function in a setting, and a management apparatus that communicates with the information processing apparatus and the function execution apparatus. The method includes generating a code corresponding to the setting with the management apparatus, transmitting the code with the management apparatus, receiving the code with the information processing apparatus, displaying the code with the information processing apparatus, reading the displayed code with the function execution apparatus, and executing the function with the function execution apparatus. The method further includes executing, with the function execution apparatus, the function based on a setting corresponding to the code read with the function execution apparatus.

[0144] There is a memory that stores a computer program which includes computer instructions. These computer instructions provide the logic and routines that enable the hardware (e.g., processing circuitry or circuitry) to perform the method disclosed herein. This computer program can be implemented in known formats as a computer-readable storage medium, a computer program product, a memory device, a record medium such as a CD-ROM or DVD, and/or the memory of an FPGA or ASIC.

Claims

1. An information processing system comprising: an information processing apparatus operated by an operator; a function execution apparatus to execute a function; and a management apparatus to communicate with the information processing apparatus and the function execution apparatus, the management apparatus including first circuitry configured to: generate a code corresponding to a setting to be used in executing the function; and transmit the code, the information processing apparatus including second circuitry configured to: receive the code; and display, on a display, the code, and the function execution apparatus including third circuitry configured to: read the code displayed on the display; and execute the function based on the setting corresponding to the code read by the third circuitry.

2. An information processing system comprising: an information processing apparatus operated by an operator; a function execution apparatus to execute a function; and a management apparatus to communicate with the information processing apparatus and the function execution apparatus, the management apparatus including first circuitry configured to transmit a value corresponding to a setting to be used in executing the function, the information processing apparatus including second circuitry configured to: receive the value; generate a code corresponding to the setting using the

value; and display, on a display, the code, and the function execution apparatus including third circuitry configured to: read the code displayed on the display; and execute the function based on the setting corresponding to the code read by the third circuitry.

3. The information processing system according to claim 2, wherein: the management apparatus further includes a memory that stores information on all settings available for use in executing the function and codes corresponding to the respective settings; and the function execution apparatus further includes another memory that stores the information on all the settings and the codes corresponding to the respective settings.

4. The information processing system according to claim 1, wherein: the management apparatus further includes a first memory that stores information on all settings available for use in executing the function and codes corresponding to the respective settings; the information processing apparatus further includes a second memory that stores the code and a setting corresponding to the code; and the function execution apparatus further includes a third memory that stores the information on all the settings and the codes corresponding to the respective settings.

5. The information processing system according to claim 1, wherein: the function includes a plurality of functions having a plurality of versions; the management apparatus further includes a memory that stores all settings available for each of the plurality of versions and codes corresponding to the respective settings for each version; the function execution apparatus further includes another memory that stores all the settings available for each of the plurality of versions and the codes corresponding to the respective settings for each version; the first circuitry is configured to generate a code corresponding to one of the plurality of versions and one of the settings; and the function execution apparatus executes the function based on the one of the plurality of versions and the one of the settings corresponding to the code.

6. The information processing system according to claim 1, wherein, the third circuitry is configured to: generate another code corresponding to the setting to be used in executing the function; and display, on another display, the code, the second circuitry is configured to: read the other code displayed on the display displayed by the third circuitry of the function execution apparatus; and transmit a value corresponding to the other code read by the second circuitry, the first circuitry is configured to: receive the value transmitted by the second circuitry of the information processing apparatus; and transmit information on the setting corresponding to the value, and the information processing apparatus further includes a memory that stores the setting of the code generated and transmitted by the management apparatus, the code generated and transmitted by the management apparatus, and the other code generated by the third circuitry and read by the second circuitry of the information processing apparatus.

7. The information processing system according to claim 1, wherein the setting includes a plurality of settings used in executing the function, and the code includes a plurality of different values for the plurality of settings.

8. The information processing system according to claim 1, wherein the code includes a two-dimensional code.

9. The information processing system according to claim 1, wherein the function execution apparatus is an image forming apparatus.

10. An information processing method performed by an information processing system including an information processing apparatus operated by an operator, a function execution apparatus, and a management apparatus, the method comprising: with the management apparatus, generating a code corresponding to a setting to be used in executing a function of the function execution apparatus; transmitting the code from the management apparatus to the information processing apparatus; displaying the code on a display for the information processing apparatus; reading the displayed code with the function execution apparatus; and executing, with the function execution apparatus, the function based on the setting corresponding to the code that is read.

11. A non-transitory recording medium storing a plurality of program codes which, when executed

by one or more processors on an information processing system including an information processing apparatus operated by an operator, a function execution apparatus, and a management apparatus, causes the one or more processors to perform an information processing method, the method comprising: with the management apparatus, generating a code corresponding to a setting to be used in executing a function of the function execution apparatus; transmitting the code from the management apparatus to the information processing apparatus; displaying the code on a display for the information processing apparatus; reading the displayed code with the function execution apparatus; and executing, with the function execution apparatus, the function based on the setting corresponding to the code that is read.
